United States Patent	12416788
Kind Code	B1
Date of Patent	September 16, 2025
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Systems and methods for performing multi-zone optical detection

Abstract

Described herein are examples of systems and methods for performing multi-zone optical detection. In an example, a multi-zone optical detection system can cover multiple detecting zones, solving problems such as detecting motion/force/speed of an air click. In another example, a group of multi-zone optical detection systems can comprise multiple multi-zone optical detection systems. In use, multiple groups of multi-zone optical detection systems can be positioned at different locations.

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Family ID: **1000008486767**

Appl. No.: **19/038075**

Filed: **January 27, 2025**

Related U.S. Application Data

us-provisional-application US 63697486 20240921

Publication Classification

Int. Cl.: **G02B13/00** (20060101); **G02B3/00** (20060101)

U.S. Cl.:

CPC **G02B13/0015** (20130101); **G02B3/0062** (20130101);

Field of Classification Search

CPC: G02B (13/0015); G02B (3/0062); G02B (6/0026); G03B (17/02); G06F (1/1626); H05B (47/13)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/697,486, filed Sep. 21, 2024, which is incorporated herein by reference as if set forth herein in its entirety.

BACKGROUND
(1) In public facilities, the transmission of bacteria and viruses through contact with control buttons, knob or dial contact, keyboard buttons, and/or other input devices (collectively referred to herein as “control buttons”) pose significant concerns for public health. Studies have shown that control buttons can harbor a variety of pathogens, including bacteria such as *Staphylococcus aureus*, *Escherichia coli*, and viruses like influenza and rhinovirus. Such viruses and bacteria can survive on surfaces like control buttons for varying amounts of time, depending on factors such as temperature, humidity, and the specific microorganism involved.
(2) When individuals touch these surfaces, they can easily transfer these pathogens to their hands. Subsequent contact with the face, mouth, or eyes can lead to the introduction of these pathogens into the body, increasing the risk of infection. Furthermore, individuals can inadvertently spread these pathogens to other surfaces and individuals with which they come into contact.
(3) To minimize the risk of infection from touching control buttons or other commonly touched surfaces, medical professionals stress the importance of practicing good hand hygiene. This can include washing one's hands frequently with soap and water for at least twenty seconds, especially after touching potentially contaminated surfaces, and using hand sanitizer containing at least 60% alcohol when soap and water are not readily available. Additionally, medical professionals recommend avoiding touching one's face, especially one's eyes, nose, and mouth, to help reduce the risk of transferring infectious agents from one's hands to one's mucous membranes.
(4) However, it is not always practical to ask individuals to wash their hands as frequently as may be required to avoid any instance of a pathogen being transferred from buttons of a control panel or other commonly touched surface to an individual's hand and then to the individual's mucous membranes. For example, there may be limited access to washing stations and/or the navigation of a facility may require contact with a large number of control buttons and similar surfaces.
(5) Furthermore, there may be a safety concern during button operating, such as stuck finger or

electricity shock. Additionally, buttons under frequent use may need regular maintenance.
(6) Given the high traffic and frequent use of control buttons and input keyboards in daily life, the risk of operating safety, maintenance work, and bacterial and viral transmission through these surfaces is heightened. As a result, a need exists for improved systems and methods for avoiding the transmission of pathogens in high traffic locations.
(7) Current technology utilizes non-contact buttons or sensors to detect an approach activity, lacking a technique to solve conflict triggers when multiple non-contact buttons or sensors needs to be implemented as a group. For example, multiple non-contact buttons, controlling electronics in a room, need to keep a large distance between each other, in order to avoid false triggers when an individual intends to activate one non-contact button and not another. Therefore, a strong need exists for improved systems and methods for detecting multi-zone movement in such locations.
SUMMARY
(8) The present disclosure describes novel, non-contact systems and methods for managing multi-zone optical detection. Applying the systems and methods described herein to any suitable devices in public facilities, industrial facilities, or private facilities, as examples, can greatly reduce the risk of bacterial and/or viral transmission through contact.
(9) In one aspect, the systems described here can detect a movement, and/or other postures. For example, the systems can analyze a pattern or series of movements. In a further aspect, the systems can translate the movements into numbers, symbols, or commands. In another aspect, the systems can send signals to, or otherwise control, output interfaces or other devices, such as open/close commands to relays, and connect/disconnect commands of a 24 V DC power to remote devices. In a further illustrative example, the systems can send signals to, or otherwise control, other devices such as valves, faucets, lights, sounds, motors, or thermostats which would otherwise require contact.
(10) In another aspect, the systems can detect the speed of a movement and/or the acceleration of the movement. For example, the systems can record a triggering time of each detection zone and calculate the speed of a movement. In another example, the system can detect a movement acceleration. In a further example, the system can output signals based on different combinations of an acceleration value and a movement velocity, such as an acceleration value of 5 g-20 g after triggering two detection zones can send a signal representing a double-click movement.
(11) In a further aspect, the systems can detect movement with one detection zone covering 200 mm away, and another detection zone covering 50 mm away. In other examples, the systems can detect movement from shorter or greater distances.
(12) In one embodiment, the systems can be arranged in arrays working as a group. For example, a group of twelve multi-zone optical detection systems can be arranged in three rows and four columns, forming a non-contact keyboard, outputting signals or commands upon respective password/symbol sequences.
(13) In further embodiments, multiple groups of the systems can be located at different places. In some examples, a device can be controlled by multiple groups of the systems simultaneously. In other examples, the groups of the systems controlling the device can have different priorities such that commands received at one or more groups are given priority and/or can override commands received at one or more other groups.
(14) In other examples, the systems can support the recording of customized movements. For example, the systems can record an individual performing a particular movement or sequential symbol input, and save associated information for later detection and use by the user.
(15) In another example, the systems can record a pattern associated with an individual's movement. In some embodiments, the systems can record a password input pattern associated with an individual's finger movement. In a further illustrative example, the systems can compare a saved recording of a prior movement with a subsequent detection of a future movement. Where the subsequently detected movement substantially matches the recorded movement, a customized

reaction (e.g., closing a relay output) can be performed.
(16) Both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the examples, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are incorporated in and constitute a part of this disclosure, illustrate various embodiments and aspects of the present disclosure. In the drawings:
(2) FIG. **1** is an illustration of example hardware components comprising a multi-zone optical detection management system in accordance with this disclosure;
(3) FIG. **2** is an illustration of example hardware components comprising an optics path unit in accordance with this disclosure;
(4) FIG. **3** is an illustration of example hardware components comprising a top layer in an optics path unit in accordance with this disclosure;
(5) FIG. **4** is an illustration of example hardware components comprising a filter layer in an optics path unit in accordance with this disclosure;
(6) FIG. **5** is an illustration of example hardware components comprising a composite layer in an optics path unit in accordance with this disclosure;
(7) FIG. **6** is an illustration of example hardware components comprising a gap layer in an optics path unit in accordance with this disclosure;
(8) FIG. **7** is an illustration of example hardware components comprising a bottom layer in an optics path unit in accordance with this disclosure;
(9) FIG. **8** is an illustration of example hardware components comprising a transmit device in a photoelectric unit in accordance with this disclosure;
(10) FIG. **9** is an illustration of example hardware components comprising a transmitter in a transmit device in accordance with this disclosure;
(11) FIG. **10** is an illustration of example hardware components comprising a receiver device in a photoelectric unit in accordance with this disclosure;
(12) FIG. **11** is an illustration of example hardware components comprising a receiver in a receiver device in accordance with this disclosure;
(13) FIG. **12** is an illustration of example components of a multi-zone optical detection system in accordance with this disclosure;
(14) FIG. **13** is an illustration of example hardware components comprising a signal processing unit in accordance with this disclosure;
(15) FIG. **14** is an illustration of example hardware components comprising a control unit in accordance with this disclosure;
(16) FIG. **15** is an illustration of example hardware components comprising an interaction unit in accordance with this disclosure;
(17) FIG. **16** is an illustration of example hardware components comprising a power supply unit in accordance with this disclosure;
(18) FIG. **17** is an illustration of example hardware components comprising a remote unit in accordance with this disclosure;
(19) FIG. **18** is an illustration of example components of a multi-zone optical detection system in accordance with this disclosure;
(20) FIG. **19** is a flowchart depicting steps in an example process for managing multi-zone optical detection in accordance with this disclosure;
(21) FIG. **20** is a flowchart depicting steps in an example process for managing multi-zone optical detection in accordance with this disclosure;

(22) FIG. **21** is a flowchart depicting steps in an example process for managing multi-zone optical detection in accordance with this disclosure;
(23) FIG. **22** is a flowchart depicting steps in an example process for managing multi-zone optical detection in accordance with this disclosure;
(24) FIG. **23** is a flowchart depicting steps in an example process for managing multi-zone optical detection in accordance with this disclosure;
(25) FIG. **24** is an illustration of example components of a multi zone optical detection system in accordance with this disclosure;
(26) FIG. **25** is a flowchart depicting steps in an example process for managing multi-zone optical detection in accordance with this disclosure;
(27) FIG. **26** is an illustration of example components of a group of multi-zone optical detection systems in accordance with this disclosure;
(28) FIG. **27** is a flowchart depicting steps in an example process for managing a group of multi-zone optical detection systems in accordance with this disclosure;
(29) FIG. **28** is an illustration of example components of a group of multi-zone optical detection systems in accordance with this disclosure;
(30) FIG. **29** is an illustration of example components of a group of multi-zone optical detection systems in accordance with this disclosure;
(31) FIG. **30** is an illustration of example components of a group of multi-zone optical detection systems in accordance with this disclosure;
(32) FIG. **31** is an illustration of example components of groups of multi-zone optical detection systems in accordance with this disclosure.

DESCRIPTION OF THE EXAMPLES

(33) Reference will now be made in detail to the present examples, including examples illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

(34) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one having ordinary skill in the art to which this disclosure belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(35) It should be understood that a number of techniques and steps are disclosed. Each of these techniques and steps has individual benefits and each can also be used in conjunction with one or more, or in some cases all, of the other disclosed techniques and steps. Accordingly, for the sake of clarity, this description will refrain from repeating every possible combination of the individual steps in an unnecessary fashion. Nevertheless, the specification and claims should be read with the understanding that such combinations are entirely within the scope of the disclosure and the claims.

(36) The present disclosure is to be considered as an exemplification of the systems and methods described here and is not intended to limit the systems and methods to the specific embodiments illustrated by the figures or described below.

(37) FIG. **1** depicts a diagram of an illustrative multi-zone optical detection management system. In one embodiment, an optics path unit **110** can modify optical transmission. Photoelectric unit **120** can transmit and receive light of multiple wavelengths. Signal processing unit **130** can process optical signals between photoelectric unit **120** and control unit **140**. Control unit **140** can exchange information and/or send signals to signal processing unit **130**, database unit **150**, interact unit **160**,

remote unit **180**, and other systems. Database unit **150** can store and manage data. Interact unit **160** can be used to interact with a user. Power supply unit **170** can provide power to the system. Remote unit **180** can communicate with control unit **140** and control other devices remotely.

(38) FIG. **2** depicts a diagram of an illustrative optics path unit **110**. In one embodiment, optics path unit **110** can comprise a top layer **210**, a filter layer **220**, a composite layer **230**, a gap layer **240**, and a bottom layer **250**. In some examples, gap layer **240** can be arranged between any two layers and optics path unit **110** can comprise multiple gap layers **240**.

(39) FIG. **3** depicts a diagram of an illustrative top layer **210**. In one embodiment, a top layer **210** can comprise a display area **310**, an auxiliary area **320**, a main area **330**, and a film **340**. In some examples, display area **310** can be coupled to one or more components of interact unit **160**, such as a display device **1520**. Auxiliary area **320** can, in some embodiments, be coupled to one or more components of interact unit **160**, such as adjustment device **1510**. In some embodiments, main area **330** can comprise any suitable structure and material designed for any reasonable purpose, such as a curved S-shape porous plastic structure for improving dynamic impact energy dissipation. In further embodiments, film **340** can comprise any suitable structure and material designed for any suitable purpose, such as waterproof and dustproof composite nanomaterials for ensuring a clean-free surface.

(40) FIG. **4** depicts a diagram of an illustrative filter layer **220**. In one embodiment, a filter layer **220** can comprise an array of filter lenses **410** of different wavelengths. In some examples, filter lens **410** can comprise any suitable structure and material, such as PMMA. Each filter lens **410** can, in some embodiments, have different physical properties, such as cutoff wavelength and refractive index.

(41) FIG. **5** depicts a diagram of an illustrative composite lens **500**. In one embodiment, a composite layer **230** can comprise an array of composite lenses **500**. Each composite lens **500** can, in some embodiments, have different structures according to different light propagation requirements. In some embodiments, composite lens **500** can comprise one or more lenses **510**, one or more composite bodies **520**, and one or more mirrors **530**. In further embodiments, composite lens **500** can change light path **540** to light path **550**. In another aspect, each lens **510** can have different physical properties, such as shape, dielectric constant, and magnetic permeability. Each composite body **520** can comprise different composite materials arranged in a periodic combination of microstructures, in some examples. In other embodiments, mirrors **530** can comprise mirrors of different shape and reflectivity.

(42) FIG. **6** depicts a diagram of an illustrative gap layer **240**. In one embodiment, gap layer **240** can comprise a spacer **610** and a filling **620**. In some examples, spacer **610** can comprise any suitable structure and material, such as a vacuum honeycomb structure. Filling **620** can, in some embodiments, comprise any suitable material, such as argon.

(43) FIG. **7** depicts a diagram of an illustrative bottom layer **250**. In one embodiment, bottom layer **250** can comprise a main structure **710** and a separator **720**. In some examples, main structure **710** can comprise any suitable structure and material, such as aluminum alloy. Separator **720** can, in some embodiments, connect transmit device **800** and receiver device **1000**. In some embodiments, separator **720** can isolate each transmitter **900** of transmit device **800**, and each receiver **1100** of receiver device **1000**.

(44) FIG. **8** depicts a diagram of an illustrative transmit device **800**. In one embodiment, transmit device **800** can comprise a transmitter No. 1 **810**, a transmitter No. 2 **820**, a transmitter No. 3 **830**, and so on, up to a transmitter No. n **840**. In some examples, some transmitters can transmit light of a same wavelength and some transmitters can transmit light of different wavelengths. Transmit device **800** can, in some embodiments, comprise seven transmitters. In such embodiments, for example, transmitter No. 1 and transmitter No. 2 can transmit light of 850 nm wavelength, transmitter No. 3 and transmitter No. 4 can transmit light of 880 nm wavelength, transmitter No. 5 and transmitter No. 6 may transmit light of 940 nm wavelength, and transmitter No. 7 can transmit

light of 560 nm wavelength.

(45) FIG. **9** depicts a diagram of an illustrative transmitter **900**. In one embodiment, transmitter **900** can comprise a top encapsulation **910**, a lens **920**, a filling **930**, a core **940**, a fixing **950**, a bottom encapsulation **960**, and a connector **970**. In some examples, top encapsulation **910** can comprise any suitable material, such as a transparent resin material. Lens **920** can, in some embodiments, comprise any suitable material, such as organic glass material for guiding and improving light emitting efficiency. In some embodiments, filling **930** can comprise any suitable material, such as argon. In further embodiments, core **940** can comprise any suitable material, such as gallium arsenide. In another aspect, fixing **950** can fix lens **920**, core **940**, and connector **970** to bottom encapsulation **960**. Bottom encapsulation **960** can comprise any suitable material, such as opaque black resin, in some examples. In other embodiments, connector **970** can include any suitable number of pins made of any suitable material, such as 3 pins made of copper-plated gold material.

(46) FIG. **10** depicts a diagram of an illustrative receiver device **1000**. In one embodiment, receiver device **1000** can comprise a receiver No. 1 **1010**, a receiver No. 2 **1020**, a receiver No. 3 **1030**, and so on, up to a receiver No. n **1040**. In some examples, some receivers can receive light of a same wavelength and some receivers can receive light of different wavelengths. Receiver device **1000** can, in some embodiments, comprise four receivers. In such embodiments, for example, receiver No. 1 and receiver No. 2 can receive light of 760 nm-1100 nm wavelength, and receiver No. 3 and receiver No. 4 can receive light of 840 nm-1100 nm wavelength.

(47) FIG. **11** depicts a diagram of an illustrative receiver **1100**. In one embodiment, receiver **1100** can comprise a top encapsulation **1110**, a lens **1120**, a filling **1130**, a core **1140**, a fixing **1150**, a bottom encapsulation **1160**, and a connector **1170**. In some examples, top encapsulation **1110** can comprise any suitable material, such as a transparent resin material. Lens **1120** can, in some embodiments, comprise any suitable material, such as an organic glass material for guiding and improving light emitting efficiency. In some embodiments, filling **1130** can comprise any suitable material, such as argon. In further embodiments, core **1140** can comprise any suitable material, such as gallium phosphide. In another aspect, fixing **1150** can fix lens **1120**, core **1140**, and connector **1170** to bottom encapsulation **1160**. Bottom encapsulation **1160** can comprise any suitable material, such as opaque black resin, in some examples. In other embodiments, connector **1170** can include any suitable number of pins made of any suitable material, such as two pins made of copper-plated gold material.

(48) FIG. **12** depicts illustrative components for a multi-zone optical detection system **1200**, a detection zone **1280**, a detection zone **1285**, and a detection zone **1290**. In some embodiments, separator **720** (as depicted in FIG. **7**) can be divided into 6×6 array channels, some of which can connect to different transmitters **900**, and some of which can connect to different receivers **1100**. In some examples, three transmitters **1210** can transmit light of 850 nm wavelength, three transmitters **1220** can transmit light of 880 nm wavelength, and three transmitters **1230** can transmit light of 940 nm wavelength, each through respective channels of optics path unit **110**. In further examples, three channels connecting three receivers **1240** can have a center wavelength of 850 nm, three channels connecting three receivers **1250** can have a center wavelength of 880 nm, three channels connecting three receivers **1260** can have a center wavelength of 940 nm, and all these channels can have a bandwidth of approximately 40 nm and a transmittance over approximately 90%. Eighteen transmitters **1270** can, in some embodiments, transmit light of any suitable wavelength, such as 500 nm-570 nm. In some embodiments, eighteen channels connecting eighteen transmitters **1270** can perform any suitable processing, such as scattering and attenuation. In further embodiments, detection zone **1280** can correspond to three transmitters **1210** and three receivers **1240**, detection zone **1285** can correspond to three transmitters **1220** and three receivers **1250**, detection zone **1290** can correspond to three transmitters **1230** and three receivers **1260**. In another aspect, detection zone **1280**, detection zone **1285**, and detection zone **1290** can be adjusted by signal processing unit **130**.

(49) FIG. **13** depicts a diagram of an illustrative signal processing unit **130**. In one embodiment, signal processing unit **130** can comprise a core processing circuit **1310**, a gain control circuit **1320**, a clock circuit **1330**, an interface and communication circuit **1340**, and an auxiliary circuit **1350**. In some embodiments, core processing circuit **1310** can comprise a digital signal processor (DSP) for performing complex signal analysis and calculation tasks. In some examples, gain control circuit **1320** can adjust the gain of the amplifier, and detect and convert signals. Clock circuit **1330** can, in some embodiments, provide a stable clock signal to maintain the timing and synchronization for signal processing. In some embodiments, interface and communication circuit **1340** can comprise an analog-to-digital converter (ADC), and a communication interface. In further embodiments, auxiliary circuit **1350** can comprise any suitable safety and power supply circuits.

(50) FIG. **14** depicts a diagram of an illustrative control unit **140**. In one embodiment, control unit **140** can comprise a power supply **1410**, a clock system **1420**, a processor **1430**, a memory **1440**, an analog-to-digital converter (ADC) **1450**, an input/output interface **1460**, a driver circuit **1470**, and a communication module **1480**. In some examples, control unit **140** can comprise any suitable architecture, such as an x86 architecture, an ARM architecture, and a RISC-V architecture. Control unit **140** can, in some embodiments, adopt any suitable circuit design, such as an STM32 board or a PLC board. In some embodiments, power supply **1410** can include power positive/negative/ground pins and a power supply voltage regulator module, providing power for corresponding analog parts, reference voltage, circuits, backup area, and clock parts in control unit **140**. In further examples, clock system **1420** can include crystal oscillator pins connecting an external crystal oscillator for providing a high-precision system clock, and an internal clock for providing a timing function for synchronization and timing operations. In another aspect, processor **1430** can comprise one or more CPUs and GPUs. Memory **1440** can include a RAM (random access memory) for temporary data storage and access, and a flash memory for long-term storage of program code and data, in some examples. In further examples, analog-to-digital converter (ADC) **1450** can include sampling, quantization, encoding, and output for converting analog signals into digital signals. In other embodiments, input/output interface **1460** can include one or more relays and/or high-speed counting interfaces. In a further aspect, driver circuit **1470** can include circuits with multiple functions, such as a display screen driver circuit or motor driver circuits. In a still further aspect, communication module **1480** can include interfaces supporting any suitable protocols, such as a serial port interface, a USB interface, a CAN interface, an Ethernet port, or a WLAN interface.

(51) In one embodiment, data unit **150** can include data storage hardware for storing and running a database. In some examples, data unit **150** can include a CPU, a memory, a large-capacity storage device, an input/output device, and an external device.

(52) FIG. **15** depicts a diagram of an illustrative interact unit **160**. In one embodiment, interact unit **160** can comprise an adjustment device **1510**, a display device **1520**, an acoustic device **1530**, and an acceleration acquisition device **1540**. In some examples, adjustment device **1510** can include buttons, switches, and a rheostat, for setting detection modes, wavelength, and effective distance of one or more corresponding detection zones. Display device **1520** can, in some embodiments, include any suitable display device, such as a display screen or display lights. In further embodiments, acoustic device **1530** can include acoustic wave generators and acoustic wave receivers. In another aspect, acceleration acquisition device **1540** can include three mutually perpendicular axial collectors for collecting vibration accelerations associated with impact forces on multi-zone optical detection system **100**.

(53) FIG. **16** depicts a diagram of an illustrative power supply unit **170**. In some embodiments, power supply unit **170** can include a voltage converting module **1610**, a power management module **1620**, and an energy storage module **1630**. In some examples, voltage converting module **1610** can convert input voltage to different working voltages for the system. Power management module **1620** can, in some embodiments, include a voltage comparator, an undervoltage lockout circuit, a protection circuit, and a temperature control system. In further embodiments, energy

storage module **1630** can comprise an energy management system.

(54) FIG. **17** depicts a diagram of an illustrative remote unit **180**. In one embodiment, remote unit **180** can include a signal receiver **1710**, a control module **1720**, and a power module **1730**. In some examples, signal receiver **1710** can comprise a wireless receiving circuit and a wired signal receiving circuit. Control module **1720** can, in some embodiments, include a CPU, a memory, input/output interfaces, and relays. In further embodiments, power module **1730** can include a voltage converter and an energy storage device. In one aspect, remote unit **180** can be configured to operate in conjunction with other third-party devices. In some embodiments, remote unit **180** can receive wireless signals from the multi-zone optical detection system **100** via WIFI. In some examples, control module **1720** can include a switch output, an analog output, and a manual control output. In another aspect, power module **1730** can include a voltage converter, for example, to convert 110 VAC-240 VAC power to 24 VDC power.

(55) FIG. **18** depicts illustrative components for a multi-zone optical detection system **100**, and a target detection region **1810**. In some examples, target detection region **1810** may include an acoustic wave coverage region of interact unit **160**, a display coverage region of interact unit **160**, and a multi-zone detection coverage region. In further examples, optics path unit can comprise a top layer **210**, a filter layer **220**, a composite layer **230**, a gap layer **240**, and a bottom layer **250**. In other embodiments, interact unit **160** can comprise an adjustment device **1510**, a display device **1520**, an acoustic device **1530**, and an acceleration acquisition device **1540**. In further embodiments, multi-zone optical detection system **100** can include one or more remote units **180**. In another aspect, one or more remote units **180** can connect to some third-party devices, and/or some remote units **180** can be integrated in some third-party devices.

(56) FIG. **19** depicts a flowchart of an illustrative method for managing multi-zone optical detection system **100**. In some embodiments, a detection process can be initiated at step **1910**. The management system can interpret the request and make a decision at step **1920**. At step **1930**, the system can execute one or more actions based on step **1920**.

(57) FIG. **20** depicts a flowchart for describing some embodiments of step **1910** of FIG. **19** in more detail. In some examples, at step **2010**, operating modes can be selected according to a preset program logic. At step **2020**, the management system can perform monitoring according to a chosen detecting mode. At step **2030**, detected information can be compared with an algorithm. At step **2050**, diagnosis can be performed. At step **2060**, signal process unit **130** can adjust transmitting device **300** based on the diagnosis result from step **2050**. At step **2040**, information and results from step **2030** and step **2060** can be processed, screening valid detection results and generating reports.

(58) FIG. **21** depicts a flowchart for describing some embodiments of step **1920** of FIG. **19** in more detail. In some examples, at step **2110**, algorithms and parameters in database unit **150** can be synchronized for analyzing detection results. At step **2120**, features and key information can be extracted and evaluated. At step **2130**, the data can be converted into high-level logic to help make management decisions. At step **2140**, simulation comparison may be performed using real data to evaluate the current algorithm. At step **2150**, the algorithm and model can be optimized through deviation and trend analysis over a long time domain. At step **2160**, algorithms and parameters can be saved and updated. At step **2170**, control unit **140** can instruct control signals to corresponding circuits.

(59) FIG. **22** depicts a flowchart for describing some embodiments of step **1930** of FIG. **19** in more detail. In some examples, output can be generated at step **2210**. At step **2220**, output signals can be delivered through wired interface. At step **2230**, other devices can receive output signals via a wired connection or a wireless connection. At step **2240**, output signals can be transmitted to interact unit **160** via a wired connection, such as to display device **1520** via a serial communication or a high voltage signal through respective input pins. At step **2250**, output signals can be delivered through wireless interface using any suitable communication protocol, such as WIFI, Bluetooth,

and Zigbee. At step **2260**, output signals can be transmitted to remote unit **180** via a wireless connection. In some examples, remote unit **180** can be integrated to other devices.

(60) FIG. **23** depicts a flowchart for describing some embodiments of step **2050** of FIG. **20** in more detail. In one embodiment, at step **2310**, transmit device **800** can stop working, and receive device **1000** can monitor ambient light of different wavelengths, such as recording intensity patterns of different wavelength with respect to time and direction. At step **2320**, one or more suitable wavelengths can be selected to cover a respective target detection zone. At step **2330**, the management system can adjust wavelength and covering distance of respective transmitter **900** of transmit device **800**. For example, a multi-zone optical detection system **100** can include two detection zones, which face a continuous ambient noise light of 880 nm wavelength in the morning, and 940 nm wavelength in the afternoon. With collected information, 850 nm wavelength can be enabled for a detection zone covering 1 cm-10 cm distance, 940 nm wavelength can be selected for a detection zone covering 1 cm-5 cm distance in the morning, and 880 nm wavelength can be selected for the detection zone covering 1 cm-5 cm distance in the afternoon. In further examples, covering distance of respective detection zone can be adjusted by signal process unit **130**.

(61) FIG. **24** depicts an example of a control method of a multi-zone optical detection system. In one aspect, a multi-zone optical detection system **100** can include a detection zone **2410** covering 1 cm-5 cm distance, a detection zone **2410** covering 1 cm-10 cm distance, a remote unit **180**, and a hand **2430**. Coverage distance and wavelength of respective detection zone **2410** and detection zone **2420** can be adjusted as described in FIG. **23**. Multi-zone optical detection system **100** can include an acceleration acquisition device **1570** to detect acceleration in X-axis, Y-axis, and Z-axis directions. For example, when the absolute acceleration value in Z-axis is greater than the absolute acceleration value in both X-axis and Y-axis, a [tap] action can be detected by the multi-zone optical detection system. As hand **2430** air clicks, detection zone **2420** and detection **2410** can be triggered in sequence. In further examples, hand **2430** can tap on the multi-zone after air click through detection zone **2420** and detection zone **2410** in sequence.

(62) FIG. **25** depicts a flowchart of an illustrative method for managing multi-zone optical detection system **100** of FIG. **24**. In one embodiment, at step **2510**, detection zone **2420** can cover a detection distance of 1 cm-12 cm, which can be used to indicate a triggering is detected. At step **2515**, a timer **1** can start timing. At step **2520**, in some examples, during the cycle of timer **1**, detection zone **2410** is not triggered. At step **2525**, according to a program setting, control unit **140** can send command signals according to process No. 1, such as displaying green light from respective transmitters **900** of transmit device **800**. At step **2530**, during the cycle of timer **1**, detection zone **2410** can be triggered. At step **2535**, timer **2** can start timing. At step **2540**, during the cycle of timer **2**, acceleration acquisition device **1570** can detect a [tap] as described in FIG. **24**. At step **2545**, according to program setting, control unit **140** can send command signals according to process No. 2, such as turning off green light from respective transmitters **900** of transmit device **800**, displaying blue light from respective transmitters **900** of transmit device **800**, and maintaining a relay on for 3 s followed by off. At step **2550**, during the cycle of timer **2**, acceleration acquisition device **1570** can detect a [tap]. At step **2555**, timer **3** can start timing. At step **2560**, in some examples, acceleration acquisition device **1570** did not detect a [tap] within the cycle of timer **3**. At step **2565**, according to program setting, control unit **140** can send command signal according to process No. 3, such as turning on remote unit **180** through WIFI communication. At step **2570**, within the cycle of timer **3**, acceleration acquisition device **1570** can detect a [tap]. At step **2575**, according to program setting, control unit **140** can send command signal according to process No. 4, such as displaying a video through display device **1520** of interact unit **160**, and playing a voice recording through acoustic device **1530** of interact unit **160**.

(63) FIG. **26** depicts an example of a group of multi-zone optical detection systems **2600**. In one aspect, twelve multi-zone optical detection systems **100** can be arrayed in four columns along the X axis and in three rows along the Y axis, with respective detection zones **2410** covering 1 cm (X

axis)×1 cm (Y axis)×5 cm (Z axis), and detection zones **2420** covering 1 cm (X axis)×1 cm (Y axis)×10 cm (Z axis). Some units included in each multi-zone optical detection system **100** can be fused into a corresponding unit of the same type. For example, the group of multi-zone optical detection systems **2600** can comprise twelve optics path units **110**, twelve photoelectric units **120**, twelve signal processing units **130**, one control unit **140**, one database unit **150**, one interaction unit **160**, one power supply unit **170**, and any suitable number of remote units **180**. In further examples, the group of multi-zone optical detection systems **2600** can comprise twelve independent multi zone optical detection systems **100**.

(64) FIG. **27** depicts a flowchart of an illustrative method for managing a group of multi-zone optical detection systems **2600** of FIG. **24**. In one embodiment, at step **2710**, the group of multi-zone optical detection systems **2600** can keep detecting. At step **2715**, any detection zone **2420** can be triggered, but no detection zone **2410** can be triggered. At step **2720**, every multi-zone optical detection system **100** with detection zone **2420** triggered in the row of highest priority on Y axis can be selected. At step **2725**, one multi-zone optical detection system **100** with detection zone **2420** triggered of highest priority on X axis can be selected as target. At step **2730**, detection zone **2420** of the target can keep triggering, but detection zone **2410** cant not be triggered. At step **2735**, the group of multi-zone optical detection systems **2600** can execute process No. 1. At step **2740**, any triggered detection zone **2420** can stop triggering, or new detection zone **2420** can be triggered. At step **2745**, it can be detected that the target triggers ZONE **2410**. At step **2750**, the group of multi-zone optical detection systems **2600** can execute process No. 2. At step **2755**, each time a target triggers its zone **2410**, the group of multi-zone optical detection systems **2600** can determine whether to enable step **2760** only, step **2710** only, or both. At step **2760**, the group of multi-zone optical detection systems **2600** can execute process No. 3. At step **2765**, the combined systems **2600** can execute process No. 4. At step **2770**, an acceleration acquisition device **1540** can detect a [tap]. At step **2775**, timer No. 1 can start timing, and the process can go back to step **2710** in case a second [tap] action is not detected until timer No. 1 ends. At step **2780**, before timer No. 1 ends, a second [tap] action can be detected by acceleration acquisition device **1540**. At step **2785**, the group of multi-zone optical detection systems **2600** can execute process No. 4.

(65) FIG. **28** depicts a flowchart for describing some embodiments of step **2715** of FIG. **27** in more detail. In one embodiment, multi-zone optical detection system **2820**, multi-zone optical detection system **2830**, multi-zone optical detection system **2840**, multi-zone optical detection system **2850**, and multi-zone optical detection system **2860** can detect a hand **2810** triggering a respective detection zone **2420**. Display device **1520** can display previous information on a screen.

(66) FIG. **29** depicts a flowchart for describing some embodiments of step **2735** of FIG. **27** in more detail. In one embodiment, at step **2720**, row No. 1 can have higher priority than row No. 2. Therefore, multi-zone optical detection system **2820**, multi-zone optical detection system **2830**, and multi-zone optical detection system **2840** can be selected. For example, at step **2725**, column No. 2 can have higher priority than column No. 3 and column No. 4. Multi-zone optical detection system **2820** can be selected as a target. At step **2730**, the target's detection zone **2420** can keep triggering. At step **2735**, some transmitters **900** of transmit device **800** of multi-zone optical detection system **2820** can transmit green light indicating being selected as target. At this time, display device **1520** can still display previous information on screen.

(67) FIG. **30** depicts a flowchart for describing some embodiments of step **2760** of FIG. **27** in more detail. In one embodiment, hand **2810** can air click toward multi-zone optical detection system **2820**. At step **2745**, multi-zone optical detection system **2820** can detect a trigger of detection zone **2410**. At step **2750**, transmit device **800** of multi-zone optical detection system **2820** can stop emitting green light, and some transmitters **900** can keep emitting blue light for approximately two seconds, indicating an air-clicked motion detected. At step **2755**, a group of multi-zone optical detection systems **2600** can determine whether to proceed to step **2760** or step **2765**. For example, if each multi-zone optical detection system represents a number/letter/symbol, and a password

needs to end at symbol “#”, step **2755** can decide to proceed to step **2760** upon symbol “#” being air-clicked, otherwise proceed to step **2765**. At step **2760**, display device **1520** can display any appropriate corresponding operation prompt. At step **2765**, display device **1520** can display the number/letter/symbol represented by the target. For a password display, display device **1520** can display the current symbol following some previously displayed symbols. In further examples, the group of multi-zone optical detection systems **2600** can continue to enter step **2710** after detecting no detection zone **2420** being triggered.

(68) FIG. **31** depicts an example of a group of multi-zone optical detection systems **3110**, a group of multi-zone optical detection systems **3120**, a group of multi-zone optical detection systems **3130**, a group of multi-zone optical detection systems **3140**, a remote unit **3150**, a remote unit **3160**, and a remote unit **3170**. In one aspect, the group of multi-zone optical detection systems **3110**, the group of multi-zone optical detection systems **3120**, the group of multi-zone optical detection systems **3130**, and the group of multi-zone optical detection systems **3140** can be positioned at different locations, and communicate through LAN, WLAN, or another suitable communication protocol. In some examples, remote unit **3150**, remote unit **3160**, and remote unit **3170** can be positioned at different locations. In one example, remote unit **3150** can communicate with and be controlled by the group of multi-zone optical detection systems **3110**. In another example, remote unit **3160** can communicate with and be controlled by the group of multi-zone optical detection systems **3110**, the group of multi-zone optical detection systems **3120**, the group of multi-zone optical detection systems **3130**, and the group of multi-zone optical detection systems **3140**. In a further example, remote unit **3170** can communicate with and be controlled by the group of multi-zone optical detection systems **3130**, and the group of multi-zone optical detection systems **3140**.

(69) Other examples of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the examples disclosed herein. Though some of the described methods have been presented as a series of steps, it should be appreciated that one or more steps can occur simultaneously, in an overlapping fashion, in a different order, or a different timeframe. The order of steps presented is only illustrative of the possibilities and those steps can be executed or performed in any suitable fashion. Moreover, the various features of the examples described here are not mutually exclusive. Rather, any feature of any example described here can be incorporated into any other suitable example. It is intended that the specification and examples be considered as illustrative only, with a true scope and spirit of the disclosure being indicated by the following claims.

Claims

1. A network comprising a plurality of optical detection systems in communication with each other, each optical detection system comprising: a plurality of physical layers, including a top layer, a filter layer, a composite layer, and a bottom layer, the bottom layer comprising a separator optically isolating a plurality of optical transmitters and a plurality of optical receivers; a first detection zone associated with a first subset of the plurality of optical transmitters and a first subset of the plurality of optical receivers; and a second detection zone associated with a second subset of the plurality of optical transmitters and a second subset of the plurality of optical receivers, wherein each respective optical detection system determines at least one of motion, acceleration, or speed of an object moving from the first detection zone to the second detection zone, and wherein a first optical detection system has a higher priority than a second optical detection system, such that when movement associated with the first and second optical detection systems is detected, a first command that is associated with the first optical detection system is performed and a second command that is associated with the second optical detection system is not performed.

2. The network of claim 1, wherein the filter layer comprises an array of filter lenses, each filter

lens within the array corresponding to one of the optical transmitters or one of the optical receivers.
3. The network of claim 1, further comprising a processing unit configured to differentiate between movement detected in the first and second detection zones of a first optical detection system from movement detected in the first and second detection zones of a second optical detection system.
4. The network of claim 1, wherein the first subset of optical transmitters transmit light at a first wavelength and the second subset of optical transmitters transmit light at a second wavelength.
5. The network of claim 1, wherein the first detection zone is located in a first three-dimensional space and the second detection zone is located in a second three-dimensional space.
6. The network of claim 1, wherein the first subset of the plurality of optical transmitters is configured to transmit light of a first wavelength for a first period of time and configured to transmit light of a second wavelength for a second period of time following the first period of time.
7. A method for performing multi-zone optical detection, comprising: providing a plurality of optical detection systems, each optical detection system including a plurality of optical transmitters, a plurality of optical receivers, a first detection zone, and a second detection zone; each of the plurality of optical detection systems configured to: transmit a first wavelength of light by a first subset of the plurality of optical transmitters; detect, in the first detection zone, reflections of light associated with the first subset of the plurality of optical transmitters using a first subset of the plurality of optical receivers; transmit a second wavelength of light by a second subset of the plurality of optical transmitters; detect, in the second detection zone, reflections of light associated with the second subset of the plurality of optical transmitters using a second subset of the plurality of optical receivers; and determine a movement, acceleration, or speed associated with an object moving through the first and second detection zones, wherein a first optical detection system has a higher priority than a second optical detection system, such that when the first and second optical detection systems determine the movement, acceleration, or speed associated with the object, a first command associated with the first optical detection system is performed and a second command associated with the second optical detection system is not performed.
8. The method of claim 7, further comprising: differentiating between movement detected in the first and second detection zones of the first optical detection system from movement detected in the first and second detection zones of the second optical detection system.
9. The method of claim 7, further including: positioning each of the optical detection systems at a different physical location.
10. The method of claim 7, wherein the first detection zone is located in a first three-dimensional space and the second detection zone is located in a second three-dimensional space.
11. The method of claim 10, wherein at least a portion of the first detection zone and at least a portion of the second detection zone overlap one another.
12. An optical detection system, comprising: a plurality of physical layers, including a top layer, a filter layer, a composite layer, and a bottom layer, the bottom layer comprising a separator optically isolating a plurality of optical transmitters and a plurality of optical receivers; a first detection zone associated with a first subset of the plurality of optical transmitters and a first subset of the plurality of optical receivers; a second detection zone associated with a second subset of the plurality of optical transmitters and a second subset of the plurality of optical receivers; and a processor for determining at least one of motion, acceleration, or speed of an object moving from the first detection zone to the second detection zone, wherein the optical detection system has a higher priority than a second optical detection system, such that when the optical detection system and the second optical detection system determine the at least one of motion, acceleration, or speed of the object, a first command associated with the optical detection system is performed and a second command associated with the second optical detection system is not performed.
13. The system of claim 12, wherein the first subset of the plurality of optical transmitters transmits light in a first wavelength and the second subset of the plurality of optical transmitters transmits light in a second wavelength.

14. The system of claim 12, wherein the first detection zone is located in a first three-dimensional space and the second detection zone is located in a second three-dimensional space.

15. The system of claim 12, wherein the first subset of the plurality of optical transmitters is configured to transmit light of a first wavelength for a first period of time and configured to transmit light of a second wavelength for a second period of time following the first period of time.

16. The system of claim 12, wherein: the separator comprises an array of optically isolated cells; and each of the plurality of optical transmitters and optical receivers is positioned within a respective cell of the array.

17. The system of claim 12, wherein the first subset of the plurality of optical receivers is configured to detect light transmitted by the first subset of optical transmitters and is configured not to detect light transmitted by the second subset of optical transmitters.